

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Annexure A

Coding challenge

Total Marks:25

Code of Conduct:

I T. Karimulla / 192465046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Karimulla

1. Write a Python script that generates a star-topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.

Star Topology using WHILE loop

```
n = int(input("Enter the number of hosts: "))
```

```
switch_name = input("Enter the switch name: ")
```

```
print("\nEnter Cat6 UTP cable length for each host:")
```

```
cables = []
```

```
i = 1
```

```
while i <= n:
```

```
    length = float(input(f"Length of cable for Host {i} (meters): "))
```

```
    cables.append(length)
```

```
    i += 1
```

```
print("\n===== STAR TOPOLOGY =====\n")
```

```
print("      " + switch_name)
```

```
print("      |")
```

```
print("      Central Switch")
```

```
print("      |")
```

```
i = 1
```

```
while i <= n:
```

```
    print(f"      |")
```

```
    print(f"      -- Cat6 ({cables[i-1]} m) --> Host {i}")
```

```
    i += 1
```

```
print("\n=====")
```

```
print("Topology: Star")
```

```
print("Switch :", switch_name)
```

```
print("Hosts :", n)
```

```
print("Cable : Cat6 UTP")
```



```

A 641 Dns 1187
File Edit Shell Debug Options Window Help
Python 2.7.10 (tags/v2.7.10:Aug 19 2015, 14:15:11) [AMD64] on win32
Enter "help()" below or click "help()" above for more information.
>>>
==== RESTART: C:\Python27\Python\Python27.py ====
Test Case 1: True
Test Case 2: True
Error: VTY segment length must be 100 bytes or less.
Test Case 3: False
>>>

```

Ln 9 Col 0

3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named 'csma_log.txt'.

```

import random
import time

```

```

# Number of hosts

```

```

hosts = ["Host1", "Host2", "Host3", "Host4"]

```

```

# Open log file

```

```

log_file = open("csma_log.txt", "w")

```

```

log_file.write("CSMA/CD Simulation - Star Topology\n")

```

```

log_file.write("=====\n\n")

```

```

# Simulate 10 transmission attempts
for attempt in range(1, 11):

```

```

    # Select two hosts randomly

```

```

    host1, host2 = random.sample(hosts, 2)

```

```

    log_file.write(f"Transmission Attempt {attempt}\n")

```

```

    log_file.write(f'{host1} and {host2} are attempting to transmit.\n')

```

```

    # Simulate collision

```

```

    collision = random.choice([True, False])

```

```

    if collision:

```

```

        log_file.write("Collision detected!\n")

```

```

# Generate random back-off interval
backoff1 = random.randint(1, 10)
backoff2 = random.randint(1, 10)

log_file.write(
    f'{host1} back-off interval: {backoff1} units\n'
)
log_file.write(
    f'{host2} back-off interval: {backoff2} units\n'
)

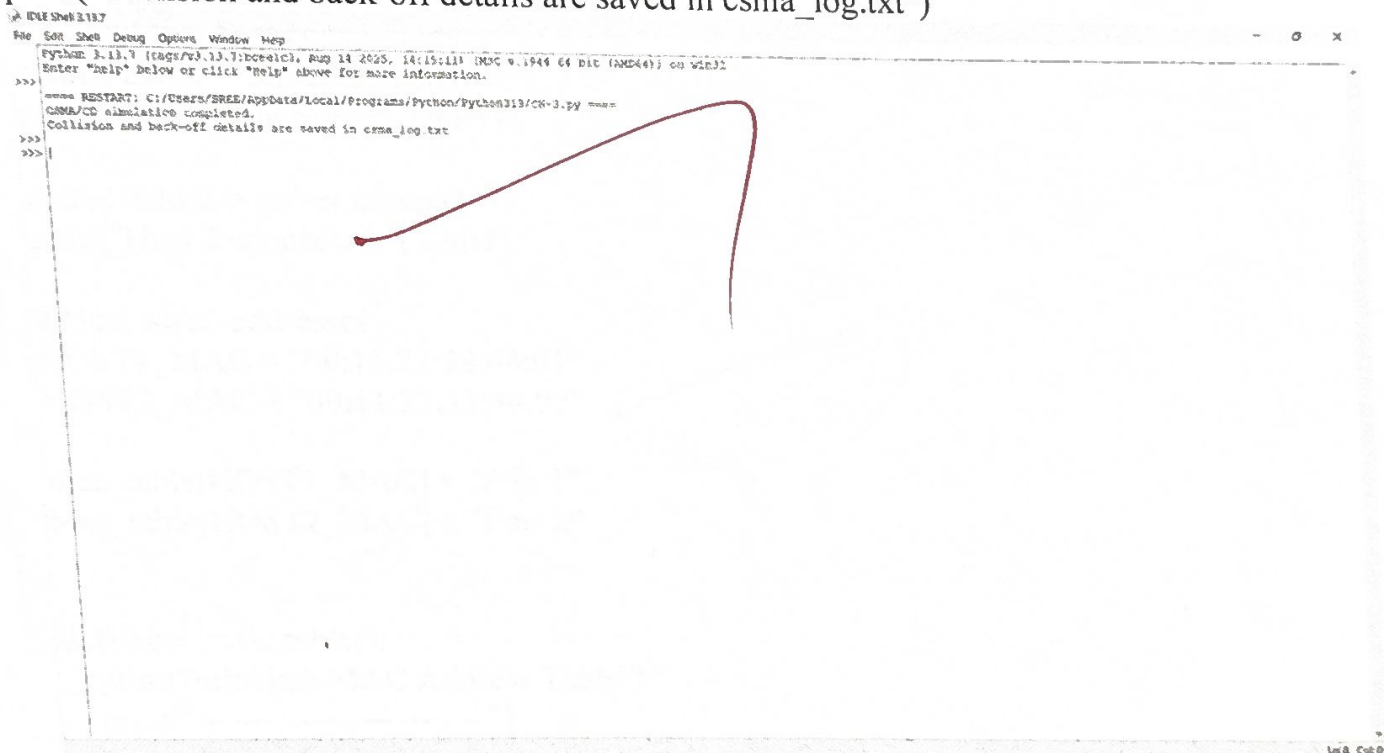
else:
    log_file.write("No collision. Transmission successful.\n")

log_file.write("-----\n")

log_file.close()

print("CSMA/CD simulation completed.")
print("Collision and back-off details are saved in csma_log.txt")

```



```

Python 3.13.7 (tags/v3.13.7:bee51c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/SREE/AppData/Local/Programs/Python/Python313/Python3.py =====
CSMA/CD simulation completed.
Collision and back-off details are saved in csma_log.txt
>>>
>>>

```

4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.

```
import socket
```

```
# Switch MAC address
```

```
SWITCH_MAC = "AA:BB:CC:DD:EE:FF"
```

```
# MAC address table
```

```
mac_table = {}
```

```
# Create TCP socket
```

```
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
server.bind(("localhost", 5000))
```

```
server.listen(2)
```

```
print("Star Switch is running...")
```

```
print("Switch MAC Address:", SWITCH_MAC)
```

```
# Accept two host connections
```

```
conn1, addr1 = server.accept()
```

```
print("Host 1 connected:", addr1)
```

```
conn2, addr2 = server.accept()
```

```
print("Host 2 connected:", addr2)
```

```
# Host MAC addresses
```

```
HOST1_MAC = "00:11:22:33:44:01"
```

```
HOST2_MAC = "00:11:22:33:44:02"
```

```
mac_table[HOST1_MAC] = "Port 1"
```

```
mac_table[HOST2_MAC] = "Port 2"
```

```
def show_mac_table():
```

```
    print("\nSwitch MAC Address Table")
```

```
    print("-----")
```

```
    for mac, port in mac_table.items():
```

```
        print(f"MAC Address: {mac} Port: {port}")
```

```
    print("-----")
```

```
# Receive message from Host 1
```

```
message = conn1.recv(1024).decode()
```

```
print("\nFrame received by switch:")
```



```

print("Source MAC:", HOST1_MAC)
print("Destination MAC:", HOST2_MAC)
print("Message:", message)

```

```

print("\nSwitch forwarding frame from Host 1 to Host 2...")
conn2.send(message.encode())

```

```

show_mac_table()

```

```

# Receive reply from Host 2
reply = conn2.recv(1024).decode()

```

```

print("\nFrame received by switch:")
print("Source MAC:", HOST2_MAC)
print("Destination MAC:", HOST1_MAC)
print("Message:", reply)

```

```

print("\nSwitch forwarding frame from Host 2 to Host 1...")
conn1.send(reply.encode())

```

```

show_mac_table()

```

```

conn1.close()
conn2.close()
server.close()

```

```

print("\nCommunication completed.")

```

```

C:\Users\USER>python 3.13.7
Python 3.13.7 (tags/v3.13.7:bee51c3, Aug 14 2025, 14:11:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python313/CR-3.py ====
CNS/CD simulation completed.
Collision and back-off details are saved in cns_log.txt
>>>
===== RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python313/CR-4.py =====
Star Switch is running...
Switch MAC Address: AA:BB:CC:DD:EE:FF

... Star Switch is running...
... Switch MAC Address: AA:BB:CC:DD:EE:FF
... Host 1 connected: ('127.0.0.1', 52341)
... Host 2 connected: ('127.0.0.1', 52342)
...
... Frame received by switch:
... Source MAC: 00:11:22:33:44:01
... Destination MAC: 00:11:22:33:44:02
... Message: Hello Host 2
...
... Switch forwarding frame from Host 1 to Host 2...
...
... Switch MAC Address Table
... MAC Address: 00:11:22:33:44:01 Port: Port 1
... MAC Address: 00:11:22:33:44:02 Port: Port 2
...
... Frame received by switch:
... Source MAC: 00:11:22:33:44:02
... Destination MAC: 00:11:22:33:44:01
... Message: Hello Host 1
...
... Switch forwarding frame from Host 2 to Host 1...
...
... Switch MAC Address Table
... MAC Address: 00:11:22:33:44:01 Port: Port 1
... MAC Address: 00:11:22:33:44:02 Port: Port 2
...
... Communication completed.

```